Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 1st Term Regular Examination, 2022

Ch 1117

(Environmental Chemistry)

Full Marks: 210 Time: 3.0 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define rate determining step (RDS). Discuss the reaction mechanism where stereochemical inversion will take place. (13)
   (b) Nucleophilic aliphatic substitution can proceed by two different mechanisms – Explain. (10)
   (c) An S_N1 reaction proceeds with racemization plus some net inversion – Explain with suitable example. (12)

2. (a) Classify polymers with different point of views. (08)
   (b) Write a step-by-step the free radical mechanism of addition polymerization by taking ethylene as an example. (08)
   (c) Write down the reactions and properties of urea-formaldehyde resins. (09)
   (d) Differentiate between Thermoplastic and Thermosetting polymers. (10)

3. (a) What do you mean by atmospheric corrosion? Write down the effect of humidity and dust particles on the rate of corrosion. (10)
   (b) "Oxygen plays a key role in the troposphere while ozone in the stratosphere" – Elucidate. (10)
   (c) How does the earth manage its radiation balance so as to maintain an average surface temperature of 15^oC? (08)
   (d) Define local action cell (LAC). How is it formed? (07)

4. (a) Explain with examples, the effect of toxic chemicals on enzymes. (07)
   (b) What is meant by water quality parameters? Describe with flow diagram the domestic waste water treatment processes. (10)
   (c) How would you broadly divide the major regions of the atmosphere? State graphically their respective altitudes and temperature ranges. What are the important chemical species in each region? (10)
   (d) Write down the essential difference between COD and BOD. (08)

Section – B

5. (a) Discuss the relative solubilities of alkaline earth metal hydroxides with proper explanation. (12)
   (b) Explain why alkali metals are highly reactive elements. (11)
   (c) Explain why transition metal elements show variable oxidation state compared to alkali and alkaline earth metals? (12)

6. (a) What product of fission is used to produce electricity? Describe the function of fuel rods, control rods, moderator and coolant in a fission reactor. (12)

(b) Define mass defect. Deduce the relationship between packing fraction and nuclear stability. (08)

(c) What is radioactive dating? The amount of carbon-14 in a piece of wood is found to be one-sixth of its amount in a fresh piece of wood. Calculate the age of old piece of wood. (08)

(d) Write down the industrial and biological applications of radio-isotopes. (07)

7. (a) Define solution. Why compounds are not a solution? (07)

(b) State Raoult's Law. How the molecular mass can be determined from vapor pressure lowering? (09)

(c) Derive an equation which correlates the elevations of boiling point with lowering of vapor pressure. (10)

(d) A stream of dry air was passed through a solution containing 2.64 g solute (an organic acid) in 30 g solvent (diethyl ether) and then through pure solvent. The loss in weight of the solution was found to be 1.29 g and that of ether was 0.069 g. Calculate the molecular weight of organic acid. (09)

8. (a) What is bond order? Write down the conditions for effective combination of atomic orbitals to form molecule. (11)

(b) Draw the Lewis dot structure and calculate the formal charges on each atom of the followings: (12)

(i) SO_4^{2-} and (ii) CO_2

(c) Draw the molecular orbital structure of nitrogen molecule and answer the followings: (12)

(i) How many bonding and antibonding electrons are in the molecule?

(ii) Explain the cause of showing magnetic properties.

(iii) Find out the bond order.

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 1st Term Regular Examination, 2022

Hum 1117

(Functional English)

Full Marks: 210 Time: 3.00 hrs.

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1 (a) Make sentences on the following structures. (14)

i. Subject + Intransitive verb + adverbial of time.
ii. Subject + Linking verb + adjective complement + extension.
iii. Subject + Linking verb + noun complement + extension.
iv. Subject + Transitive verb + gerund as object.
v. Subject + Transitive verb + object + adjective complement.
vi. Subject + Transitive verb + object + noun complement.
vii. There + verb + subject + extension.

(b) Change the following words as asked in brackets and make sentence with the changed forms: (12)

Insolvency (into adj.), Fragrant (into noun), Aggression (into verb), Accumulate (into noun), Humanity (into verb), Short (Into verb).

(c) Make a new word with each of the following prefixes and suffixes and use them in sentence: (09)

Fore…….. ; Mal…….. ; Se…….. ;
…….. ance; ……..ish; …….. ward.

2 (a) Transform the following sentences as directed. (14)

i. There is no cloud without a silver lining (Affirmative)
ii. I can never forget you. (Interrogative)
iii. Study hard and you will succeed. (Simple)
iv. Without behaving gently, you cannot win their heart. (Complex)
v. If you do not do it, you will die (Compound)
vi. A jet flies faster than an aeroplane. (Positive)
vii. It smells the sweetest of any perfume. (Comparative)

(b) Make sentences using the word as directed. (12)

Mobile (as adjective); Home (as adverb); Down (as preposition);
Either (as adjective); Either (as pronoun); Either (as conjunction).

(c) Make sentences using the following Phrases and idioms. (09)

An apple of discord; Ad hoc;
Point blank; Flesh and blood;
In a fix; Come round.

3 (a) Make WH question with underlined words of the following sentences: (14)

i. Mamun reads an interesting piece of literary works.
ii. Liza lives in a gorgeous house.
iii. Soma is 18 years old.
iv. Salim has been working for two hours.
v. Reza is driving at a high speed.
vi. The pond is waist deep.
vii. The ambiance of this palace is peaceful.

(b) Complete the sentences with subordinate clauses as directed. (12)

i. I wonder………………………………………………… (Noun clause)
ii. ………………………………………..is beyond doubt. (Noun clause)
iii. Mr. Haque is renowned professor……………….. (Adjective clause)
iv. He speaks……………………………….. (Adverb clause of Manner)
v. …………………. I would help the poor. (Adverb clause of Condition)
vi. ……………………….… , he is pious. (Adverb clause of concession)

(c) Make sentences expressing the following notions/emotions. (09)

i) Advice ii) Condolence iii) Farewell iv) Imprecation
v) Proposal vi) Suggestion

4 (a) Correct the errors of the following sentences. (14)

i. I knew why does he go.
ii. He came with a view to help me.
iii. I am still loving her.
iv. My uncle died recently.
v. I was used to smoke.
vi. He was let to go.
vii. Liza caught her hand.

(b) Make use of the following modals as asked in brackets in sentence. (12)

i. Can. (To express offer to somebody else)
ii. Can. (To express ability)
iii. Could. (To express past ability)
iv. Must. (To express certainty)
v. Dare. (To express indulgence)
vi. Would rather. (To express preference)

(c) Write two synonyms for each of the following words and make sentence with the synonyms: Bright, Courageous, Plain. (09)

Section – B

5 (a) Read the following passage carefully and answer the following questions. (20)

Pleasure and action go hand in hand; one given importance, another discarded can't indeed enliven a life-wishing to be a meaningful life, working for good for own self and for other ones in the world etc. So, Jhon Keats says pleasingly the life of the nightingale as happy – 'And with thee fade away into forest dim:' Why? It mingles a pleasure in deeds. Where? It lies in: 'Where (in the world) but to think is to be full of sorrow.' The life with pleasure only, which assesses pleasure in life without the proper business in life, can't achieve, even glimpse a lucrative life. This life explores happiness both in body and mind. So, there rises the fullness of the existence of a good financial and serene ambiance in a life. The prerequisite terms to a worthy life so finally prepare us solace as is seen in the juicy life of the protagonist of the poem – 'The Song of Thirst' by S. Das. Happy, a character in 'Death of a Salesman' by Arthur Miller leads only the life of pleasure in the world. There ultimately works no hopeful meaning in his life. The reason lies undoubtedly in the fact, he never brings any welfare for himself and other members of his family belonging to his reach. Consequentially what the world persuades us – the peoples of the world represent a wide range of meanings – achieved by action done in pleasure. All work for the inclusive well-being for the peoples etc. It, a devotive cult, rises a lived and ideological nation. W. Shakespeare rightly says: 'How far the little candle throws its beams! So shines a good deed in a naughty world'. Question:

i. How do pleasure and action move in the world? and what is the purpose?
ii. What is a lucrative life?
iii. How is the life of Happy and why is it not fruitful?
iv. What type of life does the world persuade us and why does it require so?

(b) Make a precis of the above written passage (Q.5.a) with a suitable title. (15)

6 (a) Amplify the idea – 'He is the best physician who is the most ingenious inspirer of hope'. (20)

(b) Write a paragraph on 'Exercise'. (15)

7 (a) Write a report on your departmental library. (20)

(b) Write an application for a job along with a CV. (15)

8 (a) Write a free composition on the following: "Global Co-operation: A Bright Future for the World." (35)

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BURP 1st Year 1st Term Regular Examination, 2022

Math 1117

(Mathematics - I)

Full Marks: 210 Time: 3.00 hrs

N.B. i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1 (a) Define skew-symmetric matrix, skew hermitian matrix and unitary matrix with example. (12)

(b) Show that every square matrix can be uniquely expressed as the sum of symmetric and skew-symmetric matrix. (13)

(c) Show that the matrix A $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is nilpotent matrix and find its index. (10)

2 (a) Define orthogonal matrix. If (l_r, m_r, n_r), r = 1, 2, 3 be the direction cosines of three mutually perpendicular lines referred to an orthogonal coordinates system then prove that, A = $\begin{bmatrix} l1 & m1 & n1 \\ l2 & m2 & n2 \\ l3 & m3 & n3 \end{bmatrix}$ is orthogonal. (12)

(b) Verify that $\mathrm{A.(adj\ A)} = \mathrm{(adj\ A).A} = |\mathrm{A}|\ \mathrm{I}$ where A = $\begin{bmatrix} 2 & -1 & 3 \\ 5 & 3 & 1 \\ 3 & 2 & 3 \end{bmatrix}$ (12)

(b) What is meant by singular and non-singular matrix? Using elementary row transformation, find the inverse of the matrix A (if exists). (11)

A = $\begin{bmatrix} 2 & -1 & 3 \\ 5 & 4 & -7 \\ 1 & 0 & 6 \end{bmatrix}$

3 (a) Define rank of a matrix. Find the rank of a matrix A, where A = $\begin{bmatrix} 2 & 3 & 4 \\ 1 & 5 & 7 \\ 3 & 11 & 13 \end{bmatrix}$ (12)

(b) Reduce the matrix A = $\begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix}$ to normal form and find its rank. (11)

(c) When does a system of linear equation is said to be consistent or inconsistent? Solve the system of linear equations if possible: $x + 2y + z = 2$, $2x - y + z = 1$, $3x - 4y - 4z = 16$. (12)

4 (a) Find the transformed equation of $4xy - 3y^2 - 5 = 0$ when the axes are rotated through an angle $\tan^{-1}(2)$. (12)

(b) Transform the equation $x^2 + 2xy + y^2 - 6x - 2y + 4 = 0$ to standard form and identify the conic. Also find its any two properties. (13)

(b) Explain the different type of hyperbolic function with their graph. (10)

Section – B

5 (a) Find the cartesian and spherical polar coordinates of a point whose cylindrical coordinates are $(4, \frac{\pi}{3}, 1)$. (10)

(b) Define direction cosines of a line. If l, m, n are direction cosines of a line, then prove that $l^2 + m^2 + n^2 = 1$. Find the direction cosines of the line which is perpendicular to the lines with direction cosines proportional to 1, -2, -2; 0, 2, 1. (13)

(c) A line makes angles α, β, γ, δ with the four diagonals of a cube, prove that $cos^2\alpha + cos^2\beta + cos^2\gamma + cos^2\delta = 4/3$ (12)

6 (a) A variable plane is at a consistent distance P from the origin and meets the axes in A, B and C. Show that the locus of the centroid of the tetrahedron OABC is $x^{-2} + y^{-2} + z^{-2} = 16P^{-2}$. (12)

(b) Find the distance from the point (3,4,5) to the point where the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ meet the plane $x + y + z = 2$. (10)

(c) Find the equation of the line of shortest distance between the lines $\frac{x-8}{3} = \frac{y+9}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$ (13)

7 (a) Determine the length of projection of a line segment joining the points A (x_1, y_1, z_1) and B (x_2, y_2, z_2) on the z- axis. Find the angle between the AB line and the x-axis. (12)

(b) Show that the lines $\frac{x+1}{2} = \frac{y-1}{2} = z$ and $\frac{x-1}{6} = \frac{z-3}{5} = y + 1$ are coplannar. Find their point of intersection and equation of the plane containing them. (14)

(c) Find the angle between the line $\frac{x-3}{6} = \frac{y-2}{3} = \frac{z+1}{-2}$ and the plane $2x + y + 2z + 5 = 0$. (09)

8 (a) Find the equation of the right circular cone whose vertex is at (2, -3, 5), the axis passes through (1, -5, 3), and the semi-vertical angle is $\frac{\pi}{3}$. (12)

(b) Determine the great circle distance between two cities, Dhaka ($23^0 44^/$ N, $0.24^/$ E) and Tokyo ($35^0 40^/$ N, $139^0 45^/$ E) in nautical miles. (12)

(c) Define spherical triangle. The parts of a spherical triangle are a = $56^0 40^/$, B = $96^0 47^/$ and C_A= 90^0, find the remaining parts. (11)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 1st Term Regular Examination, 2022

URP 1113

(Fundamentals of Planning Process)

Full Marks: 210 Time: 3.0 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Discuss the stages of the conventional planning process in Bangladesh with examples. (20)
   (b) What is citizen participation? Draw the ladder of citizen participation and explain the stages. (15)

2. (a) What is urban sprawl? Discuss the end results of urban sprawl of urban sprawl in metropolitan city with necessary examples. (10)
   (b) What is a primate city? Write down the advantages and disadvantages of primate cities. (10)
   (c) Briefly narrate the necessity of Urban Planning with examples in context of Bangladesh. (15)

3. (a) Write down the steps that as urban planning project (from the start to end follows). (15)
   (b) Briefly discuss the major functions of Planning Commission. (10)
   (c) "A simplified series of steps are needed in Investment Projects approval" – Briefly explain. (10)

4. (a) Annotate any five of the following: (35)
   (i) Project
   (ii) Programme
   (iii) Strategic Plan
   (iv) Structure Plan
   (v) Detailed Area Plan
   (vi) Advocacy Plan
   (vii) Land-use Plan

Section – B

5. (a) What is Urban Planning? Explain the role of Planner shaping the development of urban areas. (10)
   (b) Compare various planning theories you have learnt based on "Public Participation". (10)
   (c) Explain various factors that should be considered throughout the urban planning process. (15)

6. Assume, you have been assigned as an urban planner to a transportation related project whose goal is to ensure the safety of students at KUET road.You have three alternative idea to solve this problem. Choosing best alternative could involve conducting a survey of the students need and preferences, evaluating environmental impact, resettlement cost and other financial issues.

   Read the aforementioned statement and answer the following questions.

   (a) Using Rational Planning theory, describe the method of picking the best alternative to solve the mentioned problem. (15)

   (b) Mention some criticisms of Rational Planning. (10)

   (c) You have restricted budget for the road after selecting best alternative. How can you progress with an incremental approach then? (10)

7. (a) Describe three key components of Transactive Planning. (10)

   (b) Explain the major tasks of the Advocacy Planning Process using an example. (15)

   (c) Explain the characteristics of Radical Planning theory with an example. (10)

8. (a) Explain the Mixed Scanning Approach of planning with the help of an appropriate example. (10)

   (b) Annotate any five of the following: (25)

   (i) Rule of thumb

   (ii) Inclusive city

   (iii) Spatial Planning

   (iv) Precipitated Development

   (v) Conservation and Preservation

   (vi) Sustainable Planning

Khulna University of Engineering & Technology

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BURP1st Year1st Term Regular Examination, 2022

URP1111

(History/Evolution of Human Settlement)

Full Marks: 210 — Time: 3 hrs

N.B. i) Answer any **three** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

Section – A

1 (a) Explain the causes for the following in the stone age- (15)
i) People developed their settlements along seacoasts and mountains.
ii) Development of agriculture
iii) Development of social hierarchies

(b) Describe Stonehenge and its uses. (10)

(c) List out the Site factors and their necessity. (10)

2 (a) Describe the push and pull factors that occur in rural and urban areas. (10)

(b) Classify towns based on their functional classification and rural settlements on Patterns. (15)

(c) What are the bases of classifying a settlement as urban around the world? (10)

3 (a) Draw a "pizza slice" of the Garden City with description. (15)

(b) How did Le Corbusier plan to reduce the decongestion of the center of the city? (10)

(c) Describe the Linear City concept using Khulna as an example. (10)

4 (a) The growth and development of human settlement depended on many reasons in ancient Bengal. Find out the reasons. (14)

(b) Write short notes on: (21)
(i) Pundravarddhan, (ii) Samatata, (iii) Vanga

Section – B

5 (a) "Depending on the situation of settlement they may grow as a developed or underdeveloped" Explain with proper example. (10)

(b) Mention some features/factors that you would consider if you were alive hundreds of years ago to choose a site for a new settlement. (10)

(c) Write down the main features of Mesopotamian civilization (15)

6 (a) Mention city planning concept of Egyptian town settlement. (10)

(b) Write down the social structures of Egyptian civilization. Mention the features of Harappa city life. (15)

(c) What made the ruling period of King Hammurabi 'Golden age of Babylon'? (10)

7 (a) "The technology used in Roman construction methods is still used today" - Explain this statement. (15)

(b) Differentiate between the "Parthenon" and "Pantheon" with necessary sketch. (10)

(c) Mention the functional similarities and dissimilarities of "Ziggurats" and "Acropolis". (10)

8 (a) Why were the early Greek cities called "Humble City"? (10)

(b) "The technology used in Greek construction methods and regulations is still used today"- Explain with necessary example. (15)

(c) The function of Agora varies in different city states, why? (10)